

Tutorial 1 (Solutions)

EC3301 - Martinmas Semester

Statistical inference

Consider a random sample (iid: independently and identitically distributed) drawn from the distribution

$$y_i \sim N(2, 25)$$

1. What is the $E[y_i]$ and $Var(y_i)$?

$$E[y_i] = 2$$

$$Var(y_i) = 25$$

2. Solve for the expected value and variance of the sample mean: $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.

$$\begin{aligned} E[\bar{y}] &= E\left[\frac{1}{n} \sum_{i=1}^n y_i\right] \\ &= \frac{1}{n} \sum_{i=1}^n E[y_i] \\ &= \frac{1}{n} \sum_{i=1}^n 2 \\ &= \frac{2n}{n} \\ &= 2 \end{aligned}$$

$$\begin{aligned}
Var(\bar{y}) &= E[(\bar{y} - \underbrace{E[\bar{y}]}_{=2})^2] \\
&= E\left[\left(\frac{1}{n} \sum_{i=1}^n (y_i - 2)\right)^2\right] \\
&= E\left[\frac{1}{n^2} \sum_{i=1}^n (y_i - 2)^2 + 2 \times \frac{1}{n^2} \sum_{i=1}^{n-1} \sum_{j=i+1}^n (y_i - 2)(y_j - 2)\right] \\
&= \frac{1}{n^2} \sum_{i=1}^n \underbrace{E[(y_i - 2)^2]}_{Var(y_i)} + 2 \times \sum_{i=1}^{n-1} \sum_{j=i+1}^n \underbrace{E[(y_i - 2)(y_j - 2)]}_{Cov(y_i, y_j)=0} \\
&= \frac{n Var(y_i)}{n^2} \\
&= \frac{25}{n}
\end{aligned}$$

3. What is the distribution of $\bar{y}$?

$$\bar{y} \sim N(2, 25/n)$$

4. Compute the probability that $Pr(\bar{y} < 3)$ in a sample of $n = 100$.

Since $\bar{y} \sim N(2, 1/4)$,

$$\frac{\bar{y} - 2}{0.5} \sim N(0, 1)$$

$$\begin{aligned}
Pr(\bar{y} < 3) &= Pr(\bar{y} - 2 < 3 - 2) \\
&= Pr\left(\frac{\bar{y} - 2}{0.5} < \frac{1}{0.5}\right) \\
&= Pr(z < 2)
\end{aligned}$$

```
dis normal(2)
```

```
Running C:\Users\ndl1\websites\st-andrews-ec3301\profile.do ...
```

```
<IPython.core.display.HTML object>
```

```
.97724987
```

Suppose, you draw a random sample that has the following properties. For the next questions we will assume no knowledge of the true mean of y ; only that it the sample was drawn from a normal distribution $y_i \sim N(\mu, \sigma^2)$

```
sum y
mean y
```

Variable	Obs	Mean	Std. dev.	Min	Max
y	100	2.218156	2.082121	-2.61975	6.654652

Mean estimation

Number of obs = 100

	Mean	Std. err.	[95% conf. interval]
y	2.218156	.2082121	1.805018 2.631294

- What is the relationship between the standard deviation of the sample and the standard error of the mean reported above?

The standard deviation (s) is the square-root of the sample variance. The sample variance is an estimate of the unobserved distribution variance ($Var(y_i) = 4$). In our above example $s = 2.082121$ which is $\approx 2 = \sqrt{4}$. The standard error (se) is the square root of the $Var(\bar{y})$. As we proved above, $Var(\bar{y}) = Var(y_i)/n$. That means that $se(\bar{y}) = sd/\sqrt{n}$. As $n = 100$, we find that $se = .2082121$; exactly equal to the $s/10$.

- Earlier we showed that, $\frac{\bar{y}-2}{se(\bar{y})} \sim N(0, 1)$ where $se(\bar{y}) = \sigma/\sqrt{n}$. Since we do not know σ , what happens to the distribution if we replace σ with s , the standard deviation of the sample.

$$\frac{\bar{y} - 2}{s/\sqrt{n}} \sim T_{n-1}$$

- Using the properties of the sample, test the null hypothesis $H_0 : \mu = 1.85$ (where μ is the unobserved $E[y_i]$). Use a significance level of $\alpha = 0.05$.

$$\frac{\bar{y} - 1.85}{s/\sqrt{n}} \underset{H_0}{\sim} T_{n-1}$$

Compute the test statistic:

```
dis (_b[y]-1.85)/_se[y]
```

1.7681755

i Note

The **mean** command stores estimates in a similar way to **regress**. They are both e-class commands.

- The **_b[]** function can be used to call the stored coefficient estimate for a particular variable. The full set of coefficients is stored as in the vector **e(b)**.
- The **_se[]** function can be used to call the SE of a particular coefficient. These computed using the covariance matrix stored as **e(V)**.

To see the full list of stored estimates, type **ereturn list**. Note, this does not work for **summarize**, which is a r-class command. After summarize you can run **return list** to see the format of its stored estimates.

Rejection rule:

- Since this is a two-sided test, we will reject if $|\text{test-stat}| < t_{99,0.975}$.

The test static remains the same.

Critical values of t-distribution (which is symmetric):

```
dis invt(99,0.975)
```

1.984217

Since $|\text{test-stat}| < t_{99,0.975}$, we fail to reject.

8. Using the properties of the sample, test the null hypothesis $H_0 : \mu \leq 1.85$.

Rejection rule:

- Since this is a one-sided test, we will reject if $\text{test-stat} > t_{99,0.95}$.

Critical values of t-distribution (which is symmetric):

```
dis invt(99,0.95)
```

1.6603912

We reject H_0 and conclude that $\mu > 1.85$.

9. Why is the conclusion different for the two hypotheses?

The only difference is the critical value. The one-side test has a lower critical value than the two-sided test.

10. Which test has more statistical power? Compute the power of the test to reject H_0 when the true value of $\mu = 2$.

In the two-sided test:

$$\begin{aligned}
 Pr(\text{Reject } H_0 | \mu = 2) &= Pr\left(\left|\frac{\bar{y} - 1.85}{s/\sqrt{n}}\right| > t_{99,0.975} \mid \mu = 2\right) \\
 &= Pr\left(\frac{\bar{y} - 1.85}{s/\sqrt{n}} < t_{99,0.025} \mid \mu = 2\right) + Pr\left(\frac{\bar{y} - 1.85}{s/\sqrt{n}} > t_{99,0.975} \mid \mu = 2\right) \\
 &= Pr\left(\frac{\bar{y} - 2}{s/\sqrt{n}} < t_{99,0.025} - \frac{0.15}{s/\sqrt{n}} \mid \mu = 2\right) + Pr\left(\frac{\bar{y} - 2}{s/\sqrt{n}} > t_{99,0.975} - \frac{0.15}{s/\sqrt{n}} \mid \mu = 2\right) \\
 &= T_{n-1}(t_{99,0.025} - \frac{0.15}{s/\sqrt{n}}) + 1 - T_{n-1}(t_{99,0.975} - \frac{0.15}{s/\sqrt{n}})
 \end{aligned}$$

```
dis t(99,invtt(99,0.025)-0.15/_se[y]) + 1 - t(99,invtt(99,0.975)-0.15/_se[y])
```

.10866073

In the one-sided test:

$$\begin{aligned}
 Pr(\text{Reject } H_0 | \mu = 2) &= Pr\left(\frac{\bar{y} - 1.85}{s/\sqrt{n}} > t_{99,0.95} \mid \mu = 2\right) \\
 &= Pr\left(\frac{\bar{y} - 2}{s/\sqrt{n}} > t_{99,0.95} - \frac{0.15}{s/\sqrt{n}} \mid \mu = 2\right) \\
 &= 1 - T_{n-1}(t_{99,0.95} - \frac{0.15}{s/\sqrt{n}})
 \end{aligned}$$

```
dis 1 - t(99,invtt(99,0.95)-0.15/_se[y])
```

.17475989

This shows that the one-sided test is a more powerful test.

OLS estimator

In each case, try to prove the following properties of the OLS estimators $(\hat{\beta}_0, \hat{\beta}_1)$ for a simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

- a. $\sum_{i=1}^n \hat{u}_i = 0$, where $\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$
- b. $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.
- c. $\sum_{i=1}^n x_i \hat{u}_i = 0$
- d. $\widehat{Cov}(x_i, \hat{u}_i) = 0$ where $\widehat{Cov}()$ is the sample covariance
- e. $\widehat{Cov}(\hat{y}_i, \hat{y}_i) = \widehat{Var}(x_i)$ where $\widehat{Var}()$ is the sample variance

Question 2

Question 3